

Track 2 Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Verdict

This automated official refresh report closes `wave3_harmonic_prior_residual_refresh`.

Decision

- `wave3_harmonic_prior_residual_registry` is closed as verified exploratory baseline; not promoted over the accepted direction-parallel leaders.
- The strongest refreshed aggregate candidate is `wave3_harmonic_prior_residual_pointwise_control_Bw`.
- The accepted direction-parallel baseline changes only after a human closure review records that promotion explicitly.
- This launcher-generated report is part of the same operator run as the matrix, collage, overlay, and PDF exports.

Source Package

This official report consolidates these refreshed artifacts:

- metric matrix:
`doc/reports/analysis/track2/Track 2 Directional Model Comparison.md` ;
- matrix summary:
`output/validation_checks/track2_reference_comparison/2026-06-15-20-25-36__track2_full_directional_family_matrix_wave3_harmonic_prior_residual_track2_refresh_2026_06_15/validation_summary.yaml` ;
- per-condition metrics:
`output/validation_checks/track2_reference_comparison/2026-06-15-20-25-36__track2_full_directional_family_matrix_wave3_harmonic_prior_residual_track2_refresh_2026_06_15/per_condition_metrics.csv` ;
- best-model collage report:
`doc/reports/analysis/track2/best_model_collage_report/[2026-06-15]/track2_best_model_collage_report.md` ;
- multi-model curve comparison report:
`doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-06-15]/track2_multi_model_curve_comparison_report.md` ;
- operator launch logs:
`output/validation_checks/track2_operator_launch_logs/2026-06-15-20-25-26_wave3_harmonic_prior_residual_track2_refresh_2026_06_15` .

Candidate Refresh

The refresh added 6 candidates from wave3_harmonic_prior_residual_registry into the official 159 -candidate matrix.

Surface	Candidate	Family
global	wave3_harmonic_prior_residual_pointwise_control_global	wave3_harmonic_prior_residual_pointwise_control
Fw	wave3_harmonic_prior_residual_pointwise_control_Fw	wave3_harmonic_prior_residual_pointwise_control
Bw	wave3_harmonic_prior_residual_pointwise_control_Bw	wave3_harmonic_prior_residual_pointwise_control
global	wave3_harmonic_prior_residual_smooth_l1_structured_global	wave3_harmonic_prior_residual_smooth_l1_structured
Fw	wave3_harmonic_prior_residual_smooth_l1_structured_Fw	wave3_harmonic_prior_residual_smooth_l1_structured
Bw	wave3_harmonic_prior_residual_smooth_l1_structured_Bw	wave3_harmonic_prior_residual_smooth_l1_structured

Refreshed Source Leaders

The table ranks the refreshed source by aggregate offline Track 2 metrics.

Surface	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
Fw	wave3_harmonic_prior_residual_pointwise_control_Fw	0.003374	0.003655	7.501	12.728
Bw	wave3_harmonic_prior_residual_pointwise_control_Bw	0.003360	0.003677	7.363	13.296
global	wave3_harmonic_prior_residual_smooth_l1_structured_global	0.003399	0.003714	7.502	12.356

Refreshed Source Leaderboard

Rank	Surface	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
1	Bw	wave3_harmonic_prior_residual_pointwise_control_Bw	0.003360	0.003677	7.363	13.296
2	Fw	wave3_harmonic_prior_residual_pointwise_control_Fw	0.003374	0.003655	7.501	12.728
3	global	wave3_harmonic_prior_residual_smooth_l1_structured_global	0.003399	0.003714	7.502	12.356

Rank	Surface	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
4	Bw	wave3_harmonic_prior_residual_smooth_l1_structured_Bw	0.003431	0.003739	7.523	13.200
5	global	wave3_harmonic_prior_residual_pointwise_control_global	0.003442	0.003755	7.597	14.110
6	Fw	wave3_harmonic_prior_residual_smooth_l1_structured_Fw	0.003514	0.003768	7.812	13.703

Current Direction Leaders

These leaders are read from the matrix direction breakdown after the refresh.

Direction	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
backward	periodic_gru_sequence_Bw	0.002392	0.002639	5.466	14.820
forward	rcim_retuned_GBM19_Fw	0.001089	0.001299	2.372	4.912

Visual Evidence

The same launcher run regenerated the visual companion reports and verified that the refreshed source appears in the visual package.

Source	Collage	Overlay Forward	Overlay Backward
wave3_harmonic_prior_residual_registry	6	2	2

Closeout Decision

Wave 3 harmonic-prior residual refresh is closed as: verified exploratory baseline; not promoted over the accepted direction-parallel leaders.

Use the Wave 3 Track 2 curve, offset, collage, and overlay evidence to decide whether to continue Wave 3, move to Wave 4, or reopen latent-state modeling.